Series of real numbers

1) Calculate the sum of the following series:

a) $\sum_{n=1}^{\infty} \frac{1}{4n^2-1}$

b) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$

c) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)\dots(n+p)} \quad p \geq 1$

a) $a_n = \frac{1}{4n^2-1} = \frac{1}{(2n-1)(2n+1)} = \frac{1}{2} \cdot \frac{2n+1-(2n-1)}{(2n-1)(2n+1)}$

$= \frac{1}{2}\left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)$

$s_n = a_1 + a_2 + \dots + a_n = \sum_{k=1}^{n} a_k = \frac{1}{2}\sum_{k=1}^{n}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$

$= \frac{1}{2}\left(\sum_{k=1}^{n} \frac{1}{2k-1} - \sum_{k=1}^{n} \frac{1}{2k+1}\right) = \frac{1}{2}\left(1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \dots + \frac{1}{2n-1} - \frac{1}{2n+1}\right)$

$= \frac{1}{2}\left(1 - \frac{1}{2n+1}\right)$

$s = \lim_{n\to\infty} s_n = \frac{1}{2} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{4n^2-1} = \frac{1}{2}$

b) $a_n = \frac{1}{(n+1)(n+2)n} = \frac{1}{2}\frac{n+2-n}{n(n+1)(n+2)} = \frac{1}{2}\left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right)$

$$s_n = \sum_{k=1}^{n} a_k = \frac{1}{2}\left(\sum_{k=1}^{n} \frac{1}{k(k+1)} - \sum_{k=1}^{n} \frac{1}{(k+1)(k+2)}\right)$$

$$= \frac{1}{2}\left(\frac{1}{1\cdot 2} - \frac{1}{2\cdot 3} + \frac{1}{2\cdot 3} - \frac{1}{3\cdot 4} + \ldots + \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right)$$

$$= \frac{1}{2}\left(\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right)$$

$$s = \lim_{n\to\infty} \frac{1}{2}\left(\frac{1}{2} - \frac{1}{(n+1)(n+2)}\right) = \frac{1}{4} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{a_n(n+1)(n+2)} = \frac{1}{4}$$

c) $$\frac{1}{n(n+1)\ldots(n+p)} = \frac{1}{p}\left(\frac{1}{n(n+1)\ldots(n+p-1)} - \frac{1}{(n+1)\ldots(n+p)}\right)$$

$$= \sum_{k=1}^{n} \frac{1}{k(k+1)\ldots(k+p-1)} - \frac{1}{(k+1)\ldots(k+p)}$$

$$= \frac{1}{1\cdot 2\cdot\ldots\cdot p} - \frac{1}{2\cdot 3\cdot\ldots\cdot(p+1)} - \frac{1}{2\cdot 3\cdot\ldots\cdot(p+1)}$$

$$\frac{1}{p}\left(\frac{1}{1\cdot 2\ldots p} - \frac{n!}{(n+p)!}\right) \qquad \frac{1}{p}\left(\frac{1}{p!} - \frac{n!}{(n+p)!}\right)$$

$$\frac{1}{p!} \qquad \frac{1}{p\cdot p!}$$

$$\sum_{n=1}^{\infty} \frac{1}{n\ldots(n+p)} = \frac{1}{p\cdot p!}$$

Hw: $$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+3)}$$

2) Let $(a_n)_{n\geq 1}$ $a_{n+1} = a_n^2 - a_n + 1$, $\forall n \geq 1$, $a_1 > 1$

Prove that $\lim_{n\to\infty} a_n = +\infty$ and find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{a_n}$

If $\sum_{n=1}^{\infty} a_n$ is convergent $\Rightarrow \lim_{n\to\infty} a_n = 0$

(Criteriul de divergență)

If the $\lim_{n\to\infty} a_n \neq 0 \Rightarrow \sum_{k=1}^{n} a_n$ is divergent

ex. $\sum_{n=1}^{\infty} \frac{1}{n}$ divergent $= \infty$

$\lim_{n\to\infty} \frac{1}{n} = 0$

$\Rightarrow$ Monotonicity: $a_{n+1} - a_n = a_n^2 - 2a_n + 1$

$= (a_n - 1)^2 > 0$

$\Rightarrow a_n$ is strictly increasing $\Rightarrow a_{n+1} > a_n \Rightarrow$

$\exists \ell = \lim_{n\to\infty} a_n \in \overline{\mathbb{R}}$

We suppose $\ell \in \mathbb{R}$ and we test with the limit in the reccurance relation

$\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} (a_n^2 - a_n + 1)$

$\ell = \ell^2 - \ell + 1 \iff \ell^2 - 2\ell + 1 = 0 \iff (\ell-1)^2 = 0$

$\Rightarrow \ell = 1$ $\Big|\Rightarrow$ $\ell \neq 1$, False $\Rightarrow \lim_{n\to\infty} a_n < +\infty$

$a_1 > 1$

$a_{n+1} > a_n$

$$\sum_{n=1}^{\infty} \frac{1}{a_n} = \frac{1}{a_{n+1}-1} = \frac{1}{a_n(a_n-1)}$$

$a_{n+1} = a_n^2 - a_n + 1$ $\quad = \frac{1}{a_n - 1} - \frac{1}{a_n}$

$a_{n+1} - 1 = a_n(a_n - 1)$

$$\Rightarrow \frac{1}{a_n} = \frac{1}{a_n - 1} - \frac{1}{a_{n+1} - 1}$$

$$s_n = \sum_{k=1}^{n} \frac{1}{a_k} = \sum_{k=1}^{n} \frac{1}{a_k - 1} - \frac{1}{a_{k+1} - 1} = \frac{1}{a_1 - 1} - \frac{1}{a_{n+1} - 1}$$

$$s = \lim_{n\to\infty} s_n = \frac{1}{a_1 - 1} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{a_n} = \frac{1}{a_1 - 1}$$

3) Calculate the sum of the following series:

a) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ ⟶ alternating harmonic series

b) $\sum_{n=1}^{\infty} \frac{n^3}{n!}$

a) $a_n = \frac{(-1)^{n-1}}{n}$ $\qquad s_n = ?$

(s_{2n})
(s_{2n+1}) $\Big|$ ⟶ subsequence of s_n

$$a_{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots - \frac{1}{2n}$$

$$\gamma_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n$$

$$a_{2n} = 1 + \frac{1}{3} + \dots + \frac{1}{2n-1} - \frac{1}{2}\left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right)$$

$$a_{2n} = 1 + \frac{1}{2} + \dots + \frac{1}{2n} - \left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right)$$

$$a_{2n} = \gamma_{2n} + \ln 2n - \ln n - \gamma_n = \gamma_{2n} - \gamma_n + \ln 2$$

$$\lim_{n\to\infty} a_{2n} = \gamma - \gamma + \ln 2 = \ln 2 \quad (1)$$

$$a_{2n+1} = a_{2n} + \frac{(-1)^{2n+1-1}}{2n+1} = a_{2n} + \frac{1}{2n+1}$$

$$\lim_{n\to\infty} a_{2n+1} = \ln 2 \quad (2)$$

From (1) and (2) $\Rightarrow \lim_{n\to\infty} a_n = \ln 2$

(two subseq have the same limit and the limit is unique)

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln 2$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$

b) $\sum_{n=1}^{\infty} \frac{n^3}{n!}$, $a_n = \frac{n^3}{n!} = \frac{n^2}{(n-1)!}$

$$\sum_{n=1}^{\infty} \frac{n^2-1+1}{(n-1)!} = \sum_{n=1}^{\infty} \frac{n^2-1}{(n-1)!} + \sum_{n=1}^{\infty} \frac{1}{(n-1)!}$$

For $n=1 \Rightarrow (1 + \frac{1}{1!} + \frac{1}{2!} + \dots) = e$

$$\sum_{n=1}^{\infty} \frac{(n-1)(n+1)}{(n-1)!}, \text{ For } n=1 \Rightarrow 0$$

$$\sum_{n=2}^{\infty} \frac{n+1}{(n-2)!} = \sum_{n=2}^{\infty} \frac{n-2+3}{(n-2)!} = \sum_{n=3}^{\infty} \frac{1}{(n-3)!} + \sum_{n=2}^{\infty} \frac{3}{(n-2)!}$$

$n=2 \Rightarrow 0$ $\quad = \quad e + e + 3e = 5e+3$

Hw: $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+3)}$, $a_n = \frac{1}{n(n+1)(n+3)}$

$$\frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+3} = \frac{1}{n(n+1)(n+3)} \Rightarrow A(n+1)(n+3) + nB(n+3)$$

$+ Cn(n+1) = 1$, $n=-1 \Rightarrow B = -\frac{1}{2}$

$n=0 \Rightarrow A = \frac{1}{3}$

$n=-3 \Rightarrow C = \frac{1}{6}$

$$\Rightarrow a_n = \frac{1}{3n} - \frac{1}{2(n+1)} + \frac{1}{6(n+3)} = \frac{1}{6}\left(\frac{2}{n} - \frac{3}{n+1} + \frac{1}{n+3}\right)$$

$$= \frac{1}{6}\left(2\left(\frac{1}{n} - \frac{1}{n+1}\right) - \left(\frac{1}{n+1} - \frac{1}{n+3}\right)\right)$$

$$\sum_{k=1}^{n} \frac{1}{n+1} - \frac{1}{n+3} = \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} + \dots + \frac{1}{n-1} - \frac{1}{n+1} + \frac{1}{n} - \frac{1}{n+2}$$

$$+\frac{1}{n+1}-\frac{1}{n+3} \Rightarrow a_n = \frac{1}{6}\left(2\left(1-\frac{1}{n+1}\right)-\left(\frac{1}{2}+\frac{1}{3}-\frac{1}{n+2}-\frac{1}{n+3}\right)\right)$$

$$a = \lim a_n = \frac{1}{6}\left(2-\frac{5}{6}\right) = \frac{7}{36} \Rightarrow \sum_{n=1}^{\infty} a_n = \frac{7}{36}$$